What is the mass percent of ethyl alcohol in a solution containing 62 mL of ethyl alcohol mixed into 200 mL of water? (Note: Density of ethyl alcohol = 0.8 g/mL; density of water = 1 g/mL.)

- A. 16% [5%]
- ✔ B. 20% [47%]
- C. 25% [38%]
- D. 31% [9%]

Correct
46% answered correctly

Explanation:

Percentages can be used to express the elemental composition of a molecule or the concentration of a given component within a solution mixture. However, a variety of different ratios can be chosen as the basis for calculating a percentage. When a mass ratio is used to calculate the percentage, it is called a **mass percent (% m/m)**, defined as the **component mass divided by the total mass** of the formula unit or mixture **expressed as a percentage**.

$$\% \ (m/m) = \frac{\text{component mass}}{\text{total mass}}$$

In a solution mixture, the **total mass of the solution** is the **combined mass** of the dissolved component(s) (**solute**) and the mass of the **solvent**.

$$\% \ (m/m) =$$

When the solvent (or solute) of a solution is a liquid that is measured as a volume, the **density** of each liquid must be used to find the **mass contained in each volume** prior to completing a mass percent calculation. The density of a substance specifies the mass it contains per unit of volume that it occupies.

Applying the densities given in this question, 62 mL of ethyl alcohol and 200 mL of water equate to the following amounts of mass:

$$62 \qquad \cancel{\text{mL}}$$

$$200 \qquad \cancel{\text{mL}}$$

Using these masses, the mass percent of the ethyl alcohol in the solution is calculated to be:

$$\% \ (m/m) = —$$

(Choice A) A solution containing only 48 mL of ethyl alcohol in 200 mL of water is 16% (m/m) ethyl alcohol.

(Choice C) A value of 25% results from dividing the mass of the ethyl alcohol by the mass of the water rather than by the total mass of the ethyl alcohol and water.

(Choice D) A value of 31% results from dividing the volume of the ethyl alcohol by the volume of the water rather than calculating the ratio of the ethyl alcohol mass to the total solution mass.

Educational objective:
Mass percent (% m/m) is defined as the component mass divided by the total mass of the formula unit or mixture expressed as a percentage. The total mass of a solution mixture equals the combined mass of solute and the solvent, and the density of a liquid must be used to find its mass to complete a mass percent calculation.

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